

Table 1: Prominent learning management system (LMS) platforms

LMS platform	URL
Moodle	https://moodle.org
Open edX	https://open.edx.org
Forma LMS	https://www.formalms.org
Chamilo	https://chamilo.org
Canvas LMS	https://www.instructure.com/canvas
Sakai	https://www.sakailms.org
ILIAS	https://www.ilias.de
ATutor	https://atutor.github.io

However, there is now a need for AI-based platforms for analytics of content in academics. Using these, teachers as well as students will be able to evaluate and generate high quality content for academic applications.

AI-based platforms for teaching, academics and research

A number of AI-based platforms can be used to create content for teaching and research activities. (The platforms discussed here may or may not be open source.) A few of these are described here with the respective use cases.

ChatGPT

URL: chat.openai.com

ChatGPT is a popular AI-based platform for getting answers on different subjects. In general, the content from ChatGPT is extracted in text format but there are numerous options to create sketches, diagrams, flowcharts and other types of multimedia content.

To create sketches and flow diagrams, we can generate a Mermaid script from ChatGPT.

For instance, to create a diagram related to dam engineering, we can ask ChatGPT to generate a Mermaid script for the same. ChatGPT provides the following script, which we can paste in the online platform mermaid.live for compilation to get the unique diagram.

```
graph TD
    A[Dam Engineering] --> B[Dam Body]
    A --> C[Spillway]
    A --> D[Powerhouse]
    A --> E[Intake Structure]
    A --> F[Reservoir]
    A --> G[Outlet Works]
    A --> H[Embankment]

    B --> B1[Concrete Dam]
    B --> B2[Embankment Dam]
    C --> C1[Overflow Spillway]
    C --> C2[Controlled Spillway]
```

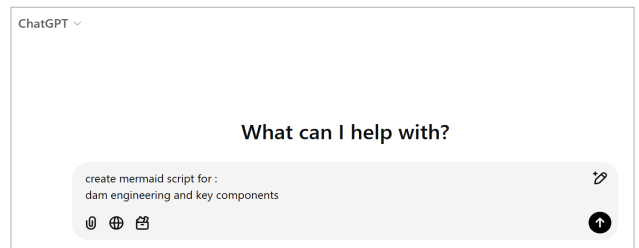


Figure 2: Writing a prompt to generate Mermaid script in ChatGPT

```
D --> D1[Generators]
D --> D2[Turbines]

E --> E1[Penstock]

F --> F1[Water Storage]

G --> G1[Release Gates]
G --> G2[Emergency Outlet]

H --> H1[Earth Dam]
H --> H2[Rockfill Dam]
```

Another diagram related to the architecture design process of a house can be generated using the following script generated by ChatGPT:

```
graph TD
    A[Architecture Design Process] --> B[Client Brief & Requirements]
    A --> C[Site Analysis]
    A --> D[Concept Design]
    A --> E[Schematic Design]
    A --> F[Design Development]
    A --> G[Construction Documentation]
    A --> H[Permits & Approvals]
    A --> I[Construction]

    B --> B1[Budget & Timeline]
    B --> B2[Function & Needs]

    C --> C1[Site Survey]
    C --> C2[Zoning & Regulations]
    C --> C3[Environmental Factors]

    D --> D1[Preliminary Floor Plan]
    D --> D2[Initial Sketches]
    D --> D3[Design Ideas]

    E --> E1[Detailed Floor Plan]
    E --> E2[3D Models & Renders]
    E --> E3[Material Selections]
```